

USER INTERFACE DESIGN

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EXPERIMENT 10

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Setting Up the HTML Form

First, create an HTML file to hold your canvas for the chart and include Chart.js.

HTML:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Inventory Visualization</title>
```

```
<style>
```

```
body {
```

```
  font-family: Arial, sans-serif;
```

```
  text-align: center;
```

```
  background-color: #f4f4f4;
```

```
}
```

```
canvas {
```

```
  margin: 20px;
```

```
  background: #fff;
```

```
  padding: 10px;
```

```
  border-radius: 8px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Inventory Management Dashboard</h1>
```

```
<canvas id="pieChart"></canvas>
```

```
<canvas id="barChart"></canvas>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Step 2: Adding JavaScript for Validation

Finally, add JavaScript to validate the input fields using Validator.js and display error messages.

JAVASCRIPT:

```
// script.js
```

```
// Inventory Data
```

```
const labels = ['Electronics', 'Clothing', 'Appliances', 'Books', 'Toys'];
```

```
const dataValues = [200, 150, 100, 80, 50];
```

```
// Common dataset
```

```
const dataset = {
```

```
  label: 'Items in Stock',
```

```
  data: dataValues,
```

```
  backgroundColor: [
```

```
'#FF6384',  
'#36A2EB',  
'#FFCE56',  
'#4BC0C0',  
'#9966FF'  
]  
};
```

// Pie Chart

```
new Chart(document.getElementById('pieChart'), {  
  type: 'pie',  
  data: {  
    labels: labels,  
    datasets: [dataset]  
  },  
  options: {  
    plugins: {  
      title: {  
        display: true,  
        text: 'Inventory Distribution'  
      }  
    }  
  }  
});
```

// Bar Chart

```
new Chart(document.getElementById('barChart'), {  
  type: 'bar',
```

```
data: {  
  labels: labels,  
  datasets: [dataset]  
},  
options: {  
  plugins: {  
    title: {  
      display: true,  
      text: 'Stock by Category'  
    }  
  },  
  
  scales: {  
    y: {  
      beginAtZero: true  
    }  
  }  
}  
});
```

OUTPUT:

Inventory Management Dashboard

Inventory Distribution

Electronics Clothing Appliances Books Toys



Stock by Category

Items in Stock

